Book Recommendation System using Machine Learning

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Abstract— Today the amount of information in the internet growth very rapidly and people need some instruments to find and access appropriate information. One of such tools is called recommendation system. Recommendation systems help to navigate quickly and receive necessary information. This paper proposes a quick and intuitive book recommendation system that helps readers to find appropriate book to read next. In this paper, implementations have been done for three algorithms i.e. Popularity based recommendation system, Content based recommendation system and Collaborative based recommendation system. This paper mainly proposes a quick book idea system that helps per clients with finding the fitting books to examine immediately.

Keywords- Recommendation System, Collaborative Filtering, Content based, Popularity Based, Music Recommendation.

INTRODUCTION

A recommender framework is a sort of data separating framework that foresee the rating or inclination that a client would provide for a thing. Recommender frameworks are at some point alluded to as suggestion framework. Recommender Systems are focused on to people who need more private experience to assess the possibly overpowering choices that a site for example might offer. Recommender frameworks frequently give customized proposal of things (to clients) utilizing a positioned rundown of anticipated things. A few organizations, for example, Amazon, Facebook and google are now are as of now using the force of recommender frameworks in building their new and existing frameworks. Most of the people are fond of reading books hence it is fascinating how every individual has different kind of book preference like novel, fiction, mystery , fantasy, poetry, and historical etc . Therefore it seems important that in order to explore different books and genres it is beneficial to have a recommendation system that can make choices for us and we don't have to waste time deciding for ourselves. A book suggestion framework is a kind of proposal framework where we need to prescribe comparable books to the peruser in view of his advantage. The reason for a book suggestion framework is to anticipate purchaser's advantage and prescribe books to them in like manner. A book suggestion framework can consider numerous boundaries like book content and book quality by separating client surveys. The books suggestion framework is utilized by online sites which give digital books like google play books, open library, great Read's, and so forth. Proposal frameworks are significant and important devices for organizations like Amazon and Netflix, who are both known for his or her customized client encounters. Our wholesome goal to create a recommendation system to help readers find their choice of book at one place so they won't face any hurdles.

Project Scope

Recommendation systems are really essential in specific endeavors as they can create an immense proportion of pay when they are capable or in like manner be a technique for standing separated in a general sense from competitors. The essential objective is to make a capable book proposition system for clients.

Target Audience

It targets on the people who loves reading and are great explorer when it comes to books as this will help them in recommending various types of books. This paper mainly highlights the importance of book recommender system and the various machine learning techniques which can be used to build such system.

Justification of the project

As stated in introduction we think there's a requirement of an advance book recommendation system. Therefore in order to achieve the aim we will use various types of machine learning recommendation techniques such as Popularity based, Content based and Collaborative based filtering that are mainly the tools for building any recommendation system After a detailed analysis of various kinds of models and their implementation we will choose the model which best suits the needs for our application.

A general architecture of a basic recommendation system is represented in Figure 1 which highlights the flow chart of working of a recommendation system. The first phase is of formation of recommendations where a refine dataset is studied and recommendations are formed using various techniques. Then the next phase is all about collecting all the recommendation and filtering them according to user's requirement. At last, the recommendation are delivered to the user.

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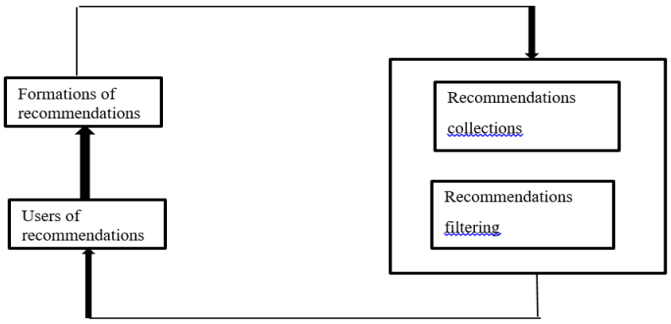


Figure 1 General Architecture

Applications of Recommendation System

These systems have their applications in various domains some of which are mentioned below:

•Retail: Retailers use the data about shoppers to have an idea about exactly what the consumer needs so that they can recommend the best to any shopper.

•E-Commerce: Companies have the data of millions of customers which is utilized in its best way for generating recommendations.

•Telecom: They have a huge amount of customer's data which leads them in generating a lot of recommendations.

RELATED WORK

This is summary of all the various research papers read during the execution period of the project in order to learn the different types of recommendation system techniques and their analysis to conclude which technique best suits our project.

AI calculations for savvy information examination and applications and has momentarily talked about how different sorts of AI techniques can be utilized for making answers for different genuine issues[1].

Further it is considered that implied criticism is more successful than other frameworks [2] and a scientific classification can help develop any recommender system..

Proposal frameworks, which consequently grasp client inclinations and make suggestions, are currently broadly utilized[3]. The proposal framework is one of the significant advances for executing personalization administrations.

Memory-based collaborative filtering (CF) has been widely followed and found to be compelling[4]. Another technique suggested further is user-user technique where the main focus is on active user neighbourhood details to throw predictions[5].

Cross filtering technique is also used for recommendation system[6].Another way is utilizing CNN technique to a build a system that can recommend items by looking at rating details and details about features. The CNN can be utilized to foresee the inert elements from the text data of the media assets. The CNN can be utilized to anticipate the dormant variables from the text data. To prepare the CNN, its feedback and result ought to be addressed first. CNN is developed to make customized proposals and accomplished a predominant outcome. The proposed calculation is checked on a public informational index. [7].

Collaborative filtering uses cosine similarity to ascertain; interest degree incorporates search times, acquiring time, and reestablishing times [8]. CBF can be used to give viable proposals.

Job recommendation algorithms is used for evaluating candidates via NLP models and to suggest jobs accordingly.

Quantum algorithm is used for recommendation wherein the past data is analyzed in order to propose recommendations. A suggestion framework utilizes the previous buys or evaluations of n items by a gathering of m clients, to give customized proposals to individual clients. The data is displayed as a m × n inclination grid which is expected to have a decent position k estimate, for a little steady k. In this work, we present a quantum calculation for proposal frameworks that has running time O(poly(k)polylog(mn)) .

Singular Value Decomposition –SVD a factorization technique used to deduct the number of data set features can be used for building a recommender system [9]. Deep Learning overshadows the other techniques because of the ability to work with non-linear data. Wide and Deep learning — mutually prepared wide direct models and profound brain organizations — to consolidate the advantages of remembrance and speculation. The creator productionized and assessed the framework on Google Play, a business portable application store with north of one billion dynamic clients and more than 1,000,000 applications. Online trial results show that Wide and Deep altogether expanded application acquisitions contrasted and wide-just and profound just models [10].DL-recommender system is better than the traditional systems as it uses techniques like CNN etc[30].

RPROP Algorithm has presented unified Boltzmann machines allowthe use of both content-based and item-specific features that allows better recommendations of both cold-start and non-cold-start items. Boltzmann machines can be utilized to join content and cooperative data to yield results that are cutthroat with cooperative method in suggesting things that have been seen previously, and furthermore viable at suggesting cold-start things .

Hybrid Algorithm is a combination of CF AND CBF algorithms that deliver a similar or most similar item as recommendation to the user .

KNN Algorithm uses item similarity to propose recommendations. KNN searches the similar items according to the metric and then generates a recommendation. In particular, the paper apply the proposed kNN method for managing meanwhile lead gathering, backslide, and missing worth attribution. This paper proposes to become familiar with a relationship framework to recreate test useful pieces of information via preparing information to dole out various k qualities to various test data of interest, alluded to as the Correlation Matrix kNN (CM-kNN for short) characterization [14].

Popularity-based is a kind of recommender framework which deals with the standard of fame or potentially anything which is in pattern [6].

Table 1 Comparison Study of Recommendation System

S.NO	DESCRIPTION	ADVANTAGES	DISADVANTAGES	ALGORITHM
[1]	This paper focusseson a far-reaching outline of AI calculations for savvy information examination and use.	This paper provides a comprehensive view on machine learning algorithms.	A more top to information is required.	Supervised, unsupervised, semi-supervised, and reinforcementlearning.
[2]	This paper is separated into five segments: proposalmethods and has given a detailed explanation.	One intriguing understanding shown is that implied criticism is more successful than express for these kindsof frameworks.	There is as yet the requirement for additional exploration that context of recommendersystems.	Content-Base, Collaborative Filtering, Demographic, Knowledge- Based and Hybrid.
[3]	This paper examines and reports the latest things, exploration opendoors in growing Recommender	Provides a generalmodel of a recommendation system.	Solutions for problems such as latency, sparsity, context-awareness problems are not provided.	Content-Based and CollaborativeFiltering.
[4]	This paper talked about the 2 conventional suggestion methods and featured their assets and difficulties.	Different learning calculations utilized in producing proposal models and execution of suggestion calculations were talked about.	.More detailed explanation about model-based and memory-based technique is required.	Model-based and Memory-Based techniques.
[5]	This papers focusses on the rich arrangement of inquiries round the client experience with the recommender.	The paper outline key challenges slowing the advance of the state of the art, and in some cases limiting the relevance of research to real- world applications.	More detailed explanation about technique is required.	User-to-user and item-to-item collaborative filtering.
[6]	This paper focusses on proposal framework and what are its sorts.	The paper has discussed CF, CBF and Hybrid filtering.	Proper details about the implementation isn't provided.	Cf, Cbf and cross filtering.
[7]	The paper proposes a substance with respect to convolutional brain organization (CNN).	CNN is developed to make customizedproposals and accomplished a predominant outcome.	The concept was slight complex to understand.	Content-put together proposal calculation based with respect to convolutional brain organization(CNN).
[8]	The target of the paper is to show different difficultiesin regards to the procedures that are being utilized for producing proposals.	The few suggestion frameworks have been suggested thatdepend on cooperative separating, content based sifting and cross breed proposal techniques.	The paper has not given a descriptive solution for various problems related to the different recommendation techniques.	Collaborative, Content, Hybrid filtering and genetic algorithm.
[9]	The paper gives an exhaustive survey of profound learning based suggestion draws near.	The paper has provided a brief knowledge of how deep learning is useful for recommender system.	Challenges such as the accuracy and scalabilityare still open for improvements and are not discussed in this paper.	CF algorithm andSVD Algorithm.
[10]	The paper presents Wide and Deep advancing mutually prepared wide direct models and profound	The paper has provided a brief knowledge about the wide and deeplearning for recommendation system.	The concept was complex to understandas the concept of wide learning is new.	Wide - DeepLearning.

	brain organizations .			
[11]	The paper proposesa characterization of semantic methodologies into hierarchical and base up.	This part presents an extensive study of semantic portrayals of thingsand client profiles.	More detail about semantic technique is required.	Content-basedfiltering and Semantic technique.
[12]	This paper gives an outline of content-based recommender frameworks.	The main concerns connected with theportrayal of things,beginning from methods for addressing organized information for unstructured data.	The shortcoming, called over-specialization, prevent these systems tobe effectively used in real world scenarios.	Content-basedfiltering.
[13]	This paper carries out client put together CF calculation with respect to a distributed computing stage, in particular Hadoop is executed, to tackle the versatility issue ofCF.	Exploratory outcomes show that a straightforward strategy that segment clients into bunches as indicated by two fundamental standards, i.e., clean game plan of mapper number tocan accomplish direct speedup.	Lack of detailed explanation.	User-based collaborative-filtering.
[14]	This article proposes to get familiar with a connection and also discussed the Correlation Matrix KNN (CM-KNN for grouping.	The paper apply the proposed KNN wayto deal with all the while lead grouping, relapse, and missing worth attribution.	The idea is slightly complex to comprehend.	KNN
[15]	This paper frames the significance of DL and presents thekinds of DL procedures and organizations.	The principal benefit of CNN is that it consequently distinguishes the huge highlights with practically no human management which made it the most utilized.	This challenge ends up being especially troublesome when continuous information handling is required.	Deep Learning

Content-based proposal frameworks prescribe things to a client by utilizing the closeness of things. This recommender framework suggests items or things in view of their portrayal or highlights. It recognizes the closeness between the items in view of their depictions. It likewise considers the user& past history to suggest a comparable item. The fundamental worries associated with the depiction of things, starting from strategies for tending to coordinated data, to extra confounding methodology coming from the Information Retrieval research region for unstructured information. Content-based recommender frameworks, fully intent on impressive a level of request on the variety of the various perspectives engaged with their plan and execution [12].

Semantic Depictions presents a broad investigation of semantic depictions of things and client profiles that undertaking to overcome the chief issues of the simpler procedures considering expressions [11].

Cooperative Filtering (CF) calculations are broadly utilized in a ton of recommender frameworks, nonetheless, the computational intricacy of CF is high in this way thwart their utilization in enormous scope frameworks [13].

Cross variety computation where we join no less than two estimations, so it helps the idea system to recommend the book considering the buyer's benefit. Book Recommendation System (BRS) considering joined components of content based filtering (CBF), agreeable isolating (CF) and association rule mining to make useful and reasonable proposition. This paper presents Book Recommendation System (BRS) in light of joined highlights of content based sifting (CBF), cooperative separating (CF) and affiliation rule mining to deliver productive and powerful advice .

Cooperative sifting utilizes cosine similitude to compute; interest degree incorporates search times, acquiring time, getting times, acquiring stretch, and recharging times as per book credits. This paper proposes a book suggestion calculation in view of cooperative separating and interest. Cooperative sifting involves cosine similitude for examination, and the interest degree involves the fundamental credits of the book as an estimation file. Through breaking down the factual information of the library of Wuxi Vocational College of Science and Technology for a long time, involving MAE and RMSE two estimation markers for trial examination, it is presumed that the technique proposed in this paper has a decent union outcome. The objective of the subsequent stage is to streamline the cooperative sifting calculation and simultaneously to upgrade the estimation markers, to have better combination results .

METHODOLOGIES

The following **Figure 2** is the methodology used to achieve the aim of building a book recommendation system.

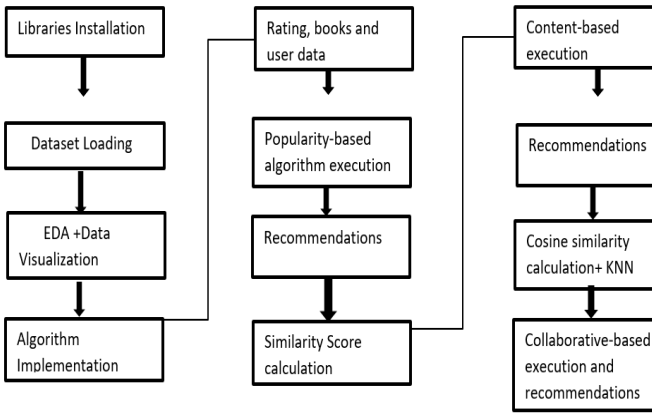


Figure 2 Methodology

•***Libraries Installation****:* Installation of following: the numpy, pandas, sklearn and matplotlib libraries. Along with this a dataset of around 2 Lakh Books data is also collected for further analysis.

•***Dataset Loading****:*Next is to store the dataset in CSV files for future use..

•***EDA+Data Visualization****:* Third step is to perform exploratory data analysis thereby retrieving the useful information from the dataset and avoiding the other details. Along with this data visualization is performed representing the data into graphs and charts etc.

•***Algorithm Implementation****:*In this step basic filtering algorithm is performed.

•***Rating,books and user data:*** In this step a structured dataset containing information about rating of the book,name of the books and reader details is maintained.

•***Popularity-based algorithm execution:*** In this step the algorithm works in such a manner that it finds ot the books with maximum rating.

•***Similarity Score Calculation:*** In this step similarity between the books is calculated or the book tat is most similar to reader's interest is presented.

•***Content based execution:*** In this step books are recommended based on the content previously referred by the reader or which the reader generally prefers.

•***Cosine similarity+KNN:*** In this similarity calculation and KNN is performed for reflecting the similarities between readers choices.

•***Collaborative-based execution:*** The cooperative sifting approach centers around finding clients who have given comparative evaluations to similar books, subsequently making a connection between clients, to whom will be proposed books that were checked on in a positive manner.

•***Recommendations:***This step basically hoghlights the suggested recommendations which we get as a result of various techniques performed during the process.

Popularity-Based Recommendation System:

It is a kind of recommender framework which deals with the guideline of fame or potentially anything which is in pattern. The **figure 3** shows books that are generally famous among the readers and straightforwardly suggest them.

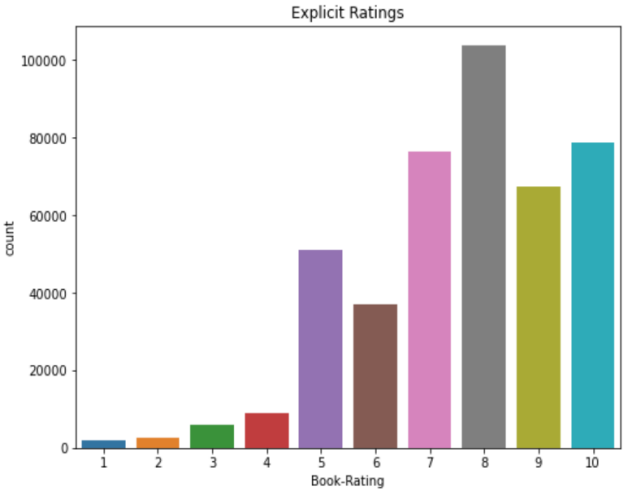


Figure 3 Popularity Based Recommendation System

For instance, on the off chance that an item is in many cases bought by the vast majority, the framework will get to realize that that item is generally well known so for each new client who just marked it, the framework will prescribe that item to that client likewise and chances turns out to be high that the new client will likewise buy that.

Executing prevalence based procedure in recommender frameworks is a typical arrangement when there is deficient data about the interests and inclinations of another client or guest in the framework, the framework can foresee things that suit the client's advantage by prescribing well known things to clients.

Content-Based Recommendation System:

This framework prescribes things to a client by taking comparability of things. This recommender framework suggests items or things in light of the depiction or elements. It distinguishes the comparability between the items in light of their depiction. It likewise considers the client's past history to suggest a comparable item.

The **figure 4** shows if a client enjoys the novel Harry Potter and the Sorcerer's Stone (Harry Potter (Paperback))then the recommender framework prescribes the client to peruse other Harry Potter's books.

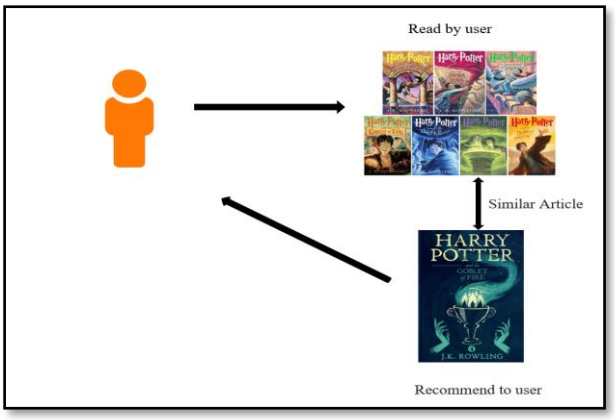


Figure 4 Content Based Recommendation System

Content-based idea structures endorse things to a client by using the similarity of things. This recommender structure proposes things or things considering their depiction or features. It recognizes the closeness between the things considering their portrayals. It similarly considers the user& previous history to recommend a near thing.

Collaborative -based Recommendation System:

It is viewed as one of the extremely brilliant recommender frameworks that work on the comparability between various clients and furthermore things that are broadly utilized as an internet business site and furthermore online film sites.

The **figure 5** shows the collaborative filtering process that centers looking around for the readers who have given comparative evaluations to similar books, thereby making a connection between readers.

This sort of proposal framework makes forecasts of what could intrigue an individual in light of the flavor of numerous different clients.

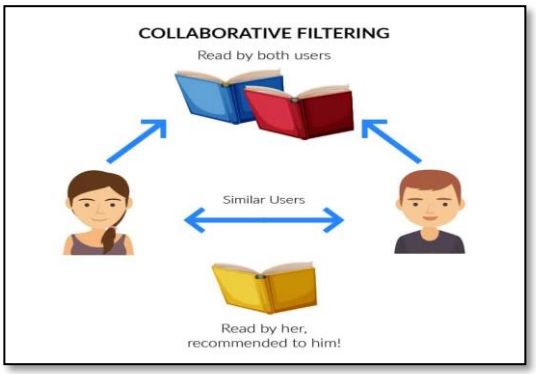


Figure 5 Collaborative Based Recommendation System

It expects to be that assuming individual X preferences Little Women book, and individual Y likes Little Women book and Black Beauty book, then, at that point, individual X could like Black Beauty book also.

IMPLEMENTAION & RESULT ANALYSIS

Dataset Used:

The dataset used for the project comprises data about 2 Lakhs books containing all the important details.

Representation of Dataset:

In **figure 6** on the y-axis we have name of the author and on the x-axis we have number of books published by the author.

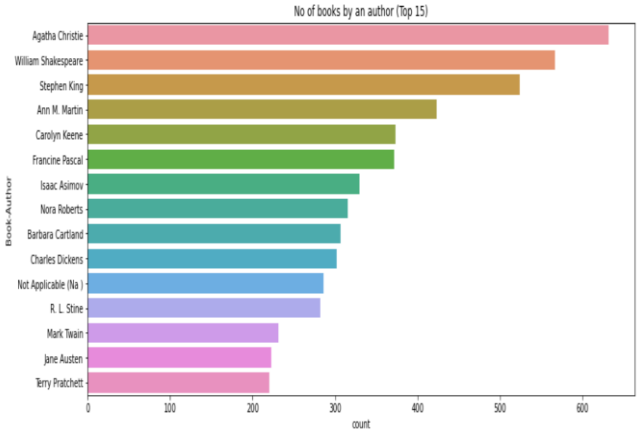


Figure 6 Dataset Representation about Author

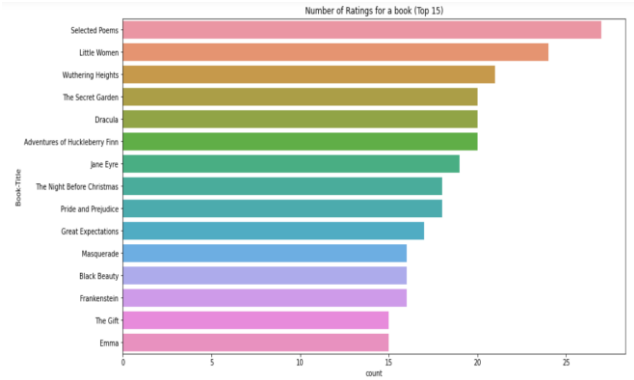


Figure 7 Dataset Representation about Books

In figure 7 we have book title on the y-axis and number of book ratings on x-axis.

Implementation & Result:

In my project I have used the following techniques for building a recommendation engine:

- Popularity-Based
- Content-Based
- Collaborative-Based

Below representing the outcomes of each of the above mentioned techniques:

1. Popularity Based:

	Book-Title	Book-Author	Image-URL-M	num_ratings	avg_rating
0	Harry Potter and the Prisoner of Azkaban (Book 3)	J. K. Rowling	http://images.amazon.com/images/P/0439136350.0...	428	5.852804
3	Harry Potter and the Goblet of Fire (Book 4)	J. K. Rowling	http://images.amazon.com/images/P/0439139597.0...	387	5.824289
5	Harry Potter and the Sorcerer's Stone (Book 1)	J. K. Rowling	http://images.amazon.com/images/P/0590353403.0...	278	5.737410
9	Harry Potter and the Order of the Phoenix (Boo...	J. K. Rowling	http://images.amazon.com/images/P/043935806X.0...	347	5.501441
13	Harry Potter and the Chamber of Secrets (Book 2)	J. K. Rowling	http://images.amazon.com/images/P/0439064872.0...	556	5.183453

```
['Book-Title'][0]
'Harry Potter and the Prisoner of Azkaban (Book 3)'
```

Figure 8 Popularity Based Technique Outcome

Figure 8 shows the most popular recommendations to the user on the basis of highest rating.

2. Content Based:

```
The 5 most recommended books to Harry Potter and the Prisoner of Azkaban (Book 3) are:

1 Harry Potter and the Chamber of Secrets (Book 2)
2 Harry Potter and the Sorcerer's Stone (Book 1)
3 Harry Potter and the Sorcerer's Stone (Harry Potter (Paperback))
4 The Nature of Good and Evil (Journey of the Soul Series, Book 3)
5 The Professor and the Madman: A Tale of Murder, Insanity, and the Making of The Oxford English Dictionary
```

Figure 9 Content Based Technique Outcome

Figure 9 shows the top 5 recommended books based on content filtering technique and recommendations specific to the user with the better accuracy than the other studied models.

3. Collabrative Based:

```
[['Animal Farm',
  'George Orwell',
  'http://images.amazon.com/images/P/0451526341.01.MZZZZZZZ.jpg'],
 ["The Handmaid's Tale",
  'Margaret Atwood',
  'http://images.amazon.com/images/P/0449212602.01.MZZZZZZZ.jpg'],
 ['Brave New World',
  'Aldous Huxley',
  'http://images.amazon.com/images/P/0060809833.01.MZZZZZZZ.jpg'],
 ['The Vampire Lestat (Vampire Chronicles, Book II)',
  'ANNE RICE',
  'http://images.amazon.com/images/P/0345313860.01.MZZZZZZZ.jpg']]
```

Figure 10 Collaborative Based Technique Outcome

Figure 10 shows the books recommended on the basis of collaborative based technique.

Result Analysis:

From **Table 1** it is clear that collaborative based recommendation is best match for the model as it shows higher accuracy and works very well large datasets which we already have. Moreover it gives a more customized recommendations.

Table 1 Comparison Table of the algorithms implemented

S No.	Proposed Approaches	Pros	Cons
1)	Popularity-Based	Helps with cold start issue	Less customized proposals
2)	Content - Based	Item cold start can be avoided	Overspecialization can occur
3)	Collaborative-Based	More customized recommendations	Security Issues

CONLUSION & FUTURE SCOPE

As you observe, the task of developing a recommendation system is not much difficult. Once we are done with preparing a goal the next step is to look for algorithms which will be used for development. In this paper we present an in view of methods, for example, notoriety based, content-based and cooperative based separating. Organizations across various areas of big business are starting to carry out suggestion frameworks trying to improve their client's web based buying experience, increment deals and hold clients. Entrepreneurs are perceiving potential in the way that proposal frameworks permit the assortment of a gigantic measure of data connecting with client's way of behaving and their exchanges inside a venture. This data can then be efficiently put away inside client profiles to be utilized for future cooperations.

Recommendation engine has a golden future ahead. Future work ought to focus on getting recommender framework information against assault and furthermore enhancing calculations.

Over the course of this subject several recommendation system techniques were performed and analyzed. After the thorough analysis it is concluded that Collaborative Based recommendation technique is best match for our book recommendation system. In coming future it shall be implemented in further project.

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